			= 1200 K	A1	[3]
			Total		**[8]**
3	**(a)**	**(i)**	(force) = $GM_1M_2/(R_1 + R_2)^2$	B1	
		(ii)	(force) = $M_1R_1\omega^2$ or $M_2R_2\omega^2$	B1	**[2]**
	(b)		ω = 2π/(1.26 x 10^8) or 2π/T	C1	
			= 4.99 x 10^{-8} rad s^{-1}	A1	**[2]**
			allow 2 s.f.: 1.59π x 10^{-8} scores 1/2		
	(c)	**(i)**	reference to either taking moments (about C) or same (centripetal) force	B1	
			$M_1R_1 = M_2R_2$ or $M_1R_1\omega^2 = M_2R_2\omega^2$	B1	
			hence $M_1/M_2 = R_2/R_1$	A0	**[2]**
		(ii)	R_2 = 3/4 x 3.2 x 10^{11} m = 2.4 x 10^{11} m	A1	
			R_1 = (3.2 x 10^{11}) – R_2 = 8.0 x 10^{10} m (allow *vice versa)*	A1	**[2]**
			if values are both wrong but have ratio of four to three, then allow 1/2		
	(d)	**(i)**	$M_2 = \{(R_1 + R_2)^2 \times R_1 \times \omega^2\} / G$ (any subject for equation)	C1	
			= (3.2 x 10^{11})^2 x 8.0 x 10^{10} x (4.99 x 10^{-8})^2/(6.67 x 10^{-11})	C1	
			= 3.06 x 10^{29} kg	A1	
		(ii)	less massive (only award this mark if reasonable attempt at **(i)**)	B1	**[4]**
			(9.17 x 10^{29} kg for more massive star)		
			Total		**[12]**
4	**(a)**		e.g. amplitude is not constant or wave is damped	B1	
			do not allow 'displacement constant'		